Lab 3, Act 2 – Answer Key

Counting detections: binomial or Poisson?

NRES 746 – Instructor Answer Key (not for student distribution)

This key follows the same section structure as the student handout. Questions with a single correct numeric answer are answered directly; open-ended items (interpretation, discussion) are given sample answers/discussion points rather than a fixed "correct" response.

Part 1: Binomial by hand

1a. The binomial's full assumption list is exactly four items – fixed N, binary (Bernoulli) outcome per trial, independence across trials, constant p across trials – and a complete answer names all four. What separates a thin answer from a strong one is engaging with "plausible or not":

- **Fixed** $N = 8$ and **binary outcome per night** are true *by construction* (the deployment length is set in advance; "at least one detection" vs. none is how the variable is defined) – not much to interrogate in either one.
- **Independence** and **constant** p are the two actually worth scrutinizing. Independence could break down if a malfunction, a multi-night weather event, or the animal revisiting a scent-marked area links one night's outcome to the next's. Constant p could drift with moon phase, temperature, food availability, or the animal habituating to (or growing wary of) the camera.

A student who only recites the four assumptions without engaging with which ones are actually shaky hasn't fully answered 1a.

1b.

```
N <- 8; p <- 0.3; k <- 3
choose(N, k) * p^k * (1 - p)^(N - k)
```

```
## [1] 0.2541218
```

```
dbinom(k, N, p)   # check
```

```
## [1] 0.2541218
```

$P(X = 3) = 0.2541$.

1c.

```
which.max(dbinom(0:8, 8, 0.3)) - 1   # mode, 0-indexed
```

```
## [1] 2
```

```
8 * 0.3                              # mean
```

```
## [1] 2.4
```

Mode = 2; mean = $Np = 2.4$. The mode can't land exactly on the mean because the mode must be a whole number (a specific count, k), while the mean is a continuous expected value with no such restriction – they coincide only in special cases (e.g. a symmetric distribution where Np happens to be an integer). Here the mean sits *above* the mode, which signals the distribution is right-skewed: bounded at 0 on the left with more room to stretch toward higher counts on the right, pulling the mean up above the single most common outcome.

Part 2: Comparing to Poisson

2a. Scenario A (binomial vs. Poisson, both mean 2.4): the two bar charts disagree most in the tails – the Poisson has a visibly fatter right tail (it has no upper bound, while the binomial is capped at 8) – and somewhat in peak height. Scenario B: the two charts are nearly indistinguishable across center, spread, and tails.

2b.

```
N <- 8; p <- 0.3
N * p * (1 - p)          # Scenario A binomial variance
```

```
## [1] 1.68
```

```
2.4                       # Scenario A/B Poisson variance (same lambda both scenarios)
```

```
## [1] 2.4
```

```
100 * 0.024 * (1 - 0.024) # Scenario B binomial variance
```

```
## [1] 2.3424
```

Scenario A: binomial variance = 1.68 vs. Poisson variance = 2.4 – mismatched. Scenario B: binomial variance = 2.3424 vs. Poisson variance = 2.4 – close match.

2c. Yes – the variance mismatch in Scenario A (1.68 vs. 2.4) explains the visible disagreement in 2a; the near-equal variances in Scenario B (2.3424 vs. 2.4) explain why those charts looked nearly identical.

2d. *Poisson is a good stand-in for binomial when N is large and p is small*, so that $(1-p) \approx 1$ and $Np(1-p) \approx Np = \lambda$. Watch for students who conclude "Poisson is just always fine" from Scenario B alone without registering that Scenario A disagreed – push them back to the variance mismatch as the reason it isn't universal.

2e. No fixed answer. Reasonable examples: a multi-night weather event (a storm suppressing detections across several consecutive nights, breaking independence), seasonal or lunar changes in animal activity (breaking constant p), or the animal habituating to (or growing wary of) the camera over the deployment (also breaking constant p).

Group discussion and revision

No fixed answer – students compare variances and their Part 2d rule of thumb, and flag their own muddiest points.